

**INTERNATIONAL CONFERENCE ON “INTEGRATING AI AND BIOMEDICAL
ENGINEERING FOR SMARTER HEALTHCARE”**

(NCABSH 2026-HYBRID MODE)

HOSTED BY

BHARATH INSTITUTE OF HIGHER EDUCATION AND RESEARCH

(Declared s deemed- to -be University under section 3 of UGC Act 1965)

TITLE

**PERFORMANCE EVALUATION AND OPTIMIZATION OF DUAL-GATE CORBON
NANO TUBE FILED EFFECT TRANSISTOR-BASED GAS SENSORS**

AUTHORS

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Mode: **Online Mode**

Session & Time: XII – 1.30pm to 4.30 pm

ABSTRACT

Field Effect Transistors (FETs) have lived to see its significant technological improvement for various applications in recent years. Carbon nanotube (CNT) based FET have been found to be a promising component for next generation transistor technologies for CNTs high carrier mobility, device stability and mechanical flexibility. However the design of CNTFET is still not well established, especially with a view to achieve the best performance still protecting thermal stability. In this study the authors had analysed the dual gate device structure and operation of transistor in which carbon nanotubes act as active channel region. DG CNTFET with different device geometrics such as channel length, oxide thickness for its output and transfer characteristics have been extensively studied using Nano TCAD ViDES simulation software. This study has thrown new insight into the device performance characteristics of DG CNTFET which can be scaled down up to minimum channel length without short channel effects.

Keywords: IC-Scaling, CNT, Nano Transistors, Flexible Electronics, Network Transistor, CNT-FET

NCABSH -2026 REGISTRATION CONFIRMATION

Abstract for Conference

Inbox x

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Srii

Apr 13, 2026, 8:01PM ☆

dear sir ,I herewith attached our research paper title " Performance Evaluation and Optimization of Dual- gate Corbon Nano Tube Filed Effect Transistor Based Ga

n

NCABSH 26

to me ▼

Apr 13, 2026, 11:14 PM ☆ 😊 ↶ ⋮

Thank you for submitting your abstract for the conference.

Thank you for your response.

You are welcome.

Thanks a lot.

↶ Reply

↷ Forward

😊

PRESENTATION @ NCABSH 2026

The screenshot shows a video conference interface with three participants: varshini, VENKAT, and Thaufiq Ahmed. The main window displays a PowerPoint presentation titled "INTRODUCTION". The slide content is as follows:

INTRODUCTION

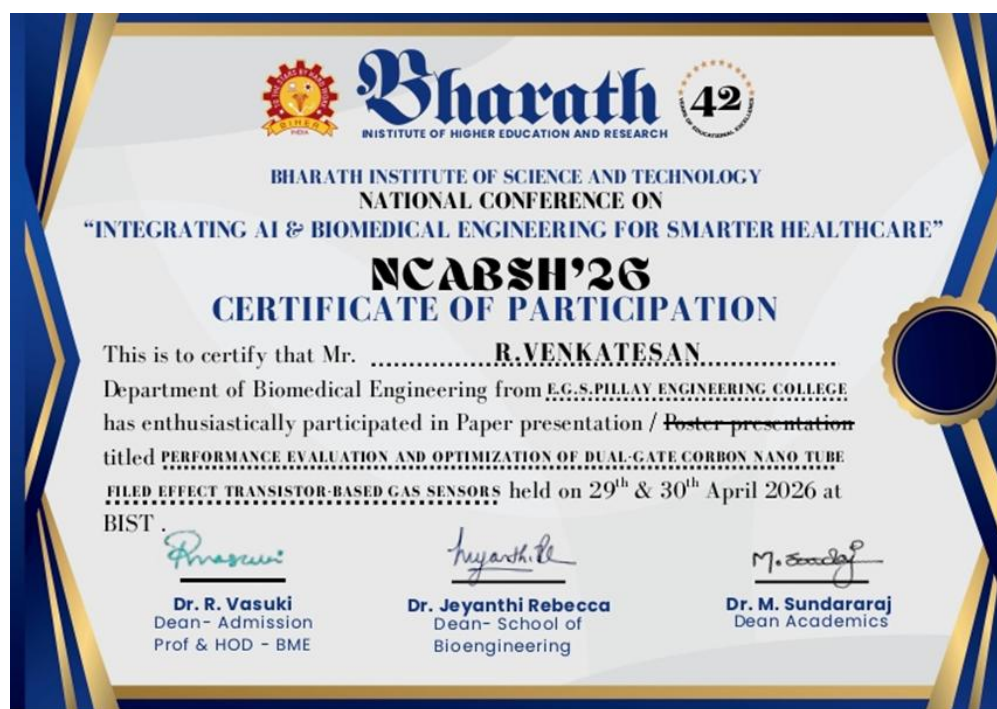
Transistors are the fundamental building blocks of modern electronic systems and are widely used in integrated circuits for switching and amplification purposes.

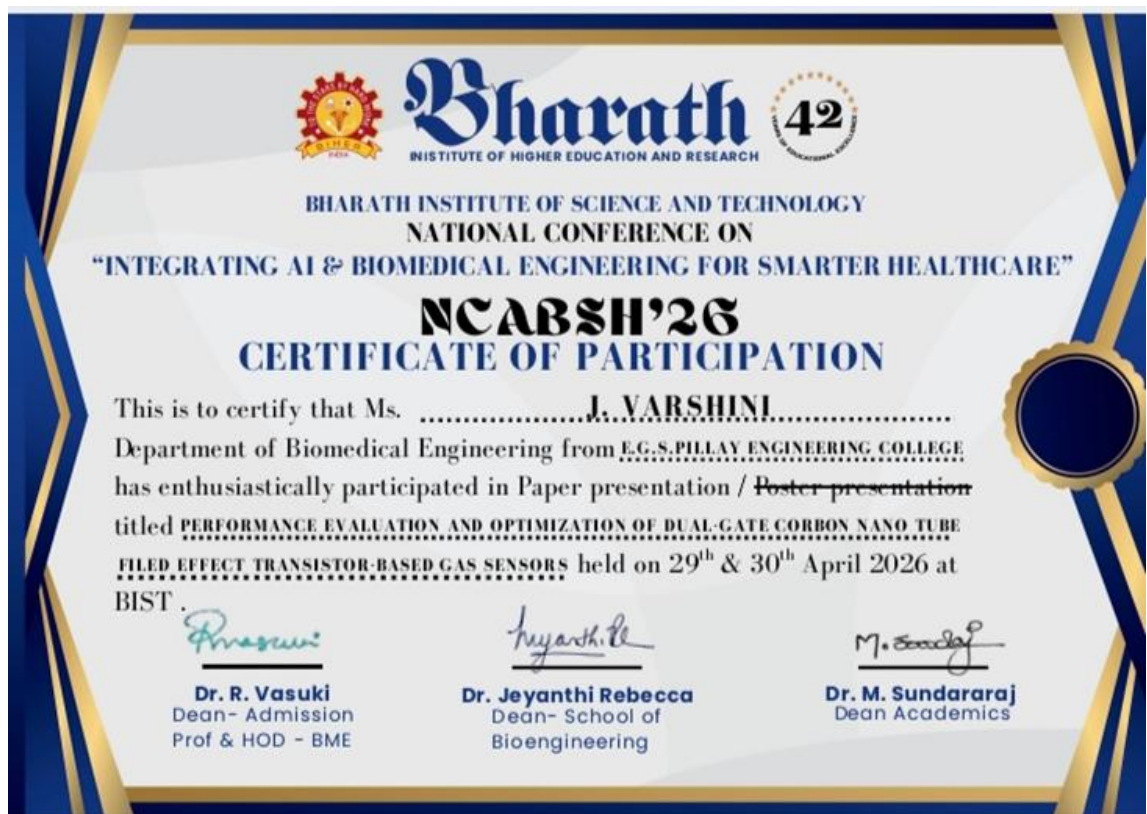
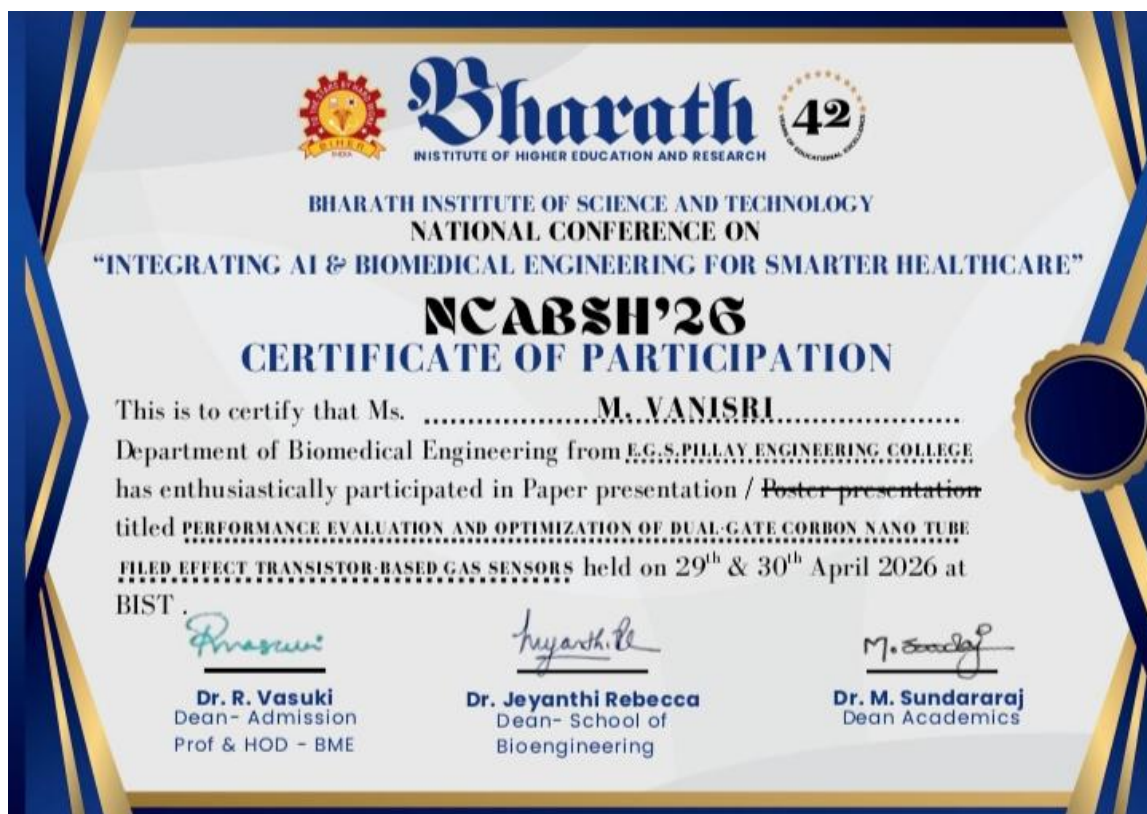
- ✓ With the continuous advancement in technology, the demand for faster, smaller, and more energy-efficient devices has increased significantly.
- ✓ This has led to the need for scaling down transistor dimensions.
- ✓ Multiple alternatives such as graphene, quantum dots and organic materials were explored for transistors

The presentation includes a sidebar with six slide thumbnails and a bottom status bar with the text "Click to add notes". A warning message at the top of the presentation area states: "Cannot detect your microphone, please check the device and connection and try again."

This screenshot provides a close-up view of the video conference window, focusing on the participant VENKAT. The interface shows the same three participants: varshini, VENKAT, and Thaufiq Ahmed, along with a fourth participant, anu loyal, whose profile picture is visible. The warning message "Cannot detect your microphone, please check the device and connection and try again." is displayed at the top of the main video area. VENKAT is wearing a yellow shirt and a lanyard with a badge that reads "AGAPATI". The background behind her shows a desk with a stack of papers and a sign on the wall that partially reads "Dr. R. V. ...".

CERTIFICATES







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42
YEARS
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NCABSH'26
CERTIFICATE OF PARTICIPATION

This is to certify that Ms. B. ITHAYA PRIYA
Department of Biomedical Engineering from E.G.S.PILLAY ENGINEERING COLLEGE
has enthusiastically participated in Paper presentation / ~~Poster presentation~~
titled PERFORMANCE EVALUATION AND OPTIMIZATION OF DUAL-GATE CARBON NANO TUBE
FIELD EFFECT TRANSISTOR-BASED GAS SENSORS held on 29th & 30th April 2026 at
BIST.

Dr. R. Vasuki
Dean- Admission
Prof & HOD - BME

Dr. Jeyanthi Rebecca
Dean- School of
Bioengineering

Dr. M. Sundararaj
Dean Academics